

9(a)

any **four** from:

4

	structure	function in respiration	
1	outer membrane	permeable for (diffusion of), oxygen / carbon dioxide / pyruvate / reduced NAD	;
2	inner membrane	site of, chemiosmosis / oxidative phosphorylation	;
3		(forms) cristae / is folded, to provide, large / increased, surface area or (has) many, electron transport chains / ETCs / electron carriers / ATP synthase (molecules)	;
4	intermembrane space	ref. to setting up proton gradient / high concentration of protons	;
5	matrix	site of (enzymes in) the, Krebs cycle / link reaction	;
6	DNA	codes for, (named) proteins / ribosomes / RNA	;

9(b)

any **three** from:

3

- 1 *idea that* more active the cell means more mitochondria ;
- 2 *idea that* heart muscle (cell) contraction needs lots of ATP ;
- 3 liver cell carries out, glycogenolysis / glycogenesis / gluconeogenesis / deamination / detoxification, so needs ATP ;
- 4 *idea that* red blood cells need room for haemoglobin ;

Question	Answer	Marks
9(c)	any three from: 1 glucose phosphorylated by ATP to form fructose (1,6-)bisphosphate ; 2 fructose (1,6-)bisphosphate splits into two molecules of, triose phosphate / TP ; R hydrolysed 3 triose phosphate / TP, oxidised to pyruvate ; A dehydrogenated 4 ATP / reduced NAD, produced ;	3

Question	Answer	Marks
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